FACULTY PROFILE

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N/A
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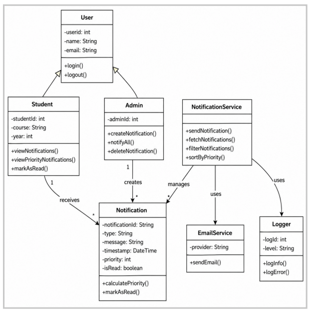

Research Area

N/A

Professional Experience

Institution / Organization	Designation	Period
National Engineering College, Kovilpatti	N/A	Till Date

Seed Money

1. IoT Based Smart Agriculture, 1 Year, 1/1/2026, 12/31/2026, 50000.00, 1 Prototype Developed

2. AI-Enabled Smart Irrigation and Crop Monitoring, 6 Months, 1/1/2025, 7/31/2025, 40000.00, Working prototype developed and initial field testing completed.

3. Low-Cost Solar Powered Water Purification System, 9 Months, 2/1/2026, 11/30/2026, 55000.00, Functional prototype developed with performance validation under controlled conditions.

4. Machine Learning Based Student Performance Prediction, 1 Year, 3/1/2025, 3/31/2026, 65000.00, Dataset prepared, predictive model developed, and accuracy evaluated using test cases.

5. IoT Enabled Energy Monitoring for Academic Buildings, 18 Months, 4/1/2026, 10/31/2027, 75000.00, Prototype monitoring unit developed with real-time data collection and dashboard visualization.

6. Smart Waste Segregation Using Computer Vision, 2 Years, 5/1/2025, 5/31/2027, 90000.00, Image dataset created, classification model trained, and a working demonstration system implemented.

7. Development of an Autonomous Agricultural Monitoring Rover, 6 Months, 6/1/2026, 12/31/2026, 50000.00, Mechanical prototype assembled, navigation algorithm tested, and controlled trials completed.

8. Deep Learning Based Early Detection of Plant Diseases, 9 Months, 7/1/2025, 4/30/2026, 85000.00, Annotated image dataset prepared and an early-stage disease detection model validated.

9. Portable Air Quality Monitoring and Alert System, 1 Year, 8/1/2026, 8/31/2027, 60000.00, Portable prototype developed with sensor calibration and real-time alert generation.

10. Blockchain Based Secure Academic Certificate Verification, 18 Months, 9/1/2025, 3/31/2027, 70000.00, Prototype verification platform developed with secure record creation and successful test transactions.

11. Smart Traffic Signal Optimization Using Real-Time Data, 2 Years, 10/1/2026, 10/31/2028, 95000.00, Simulation model developed and traffic optimization results evaluated using representative scenarios.

12. Development of a Low-Cost Assistive Device for Elderly Care, 6 Months, 11/1/2025, 5/31/2026, 45000.00, Assistive prototype designed, fabricated, and tested for basic usability and safety.

13. Renewable Energy Forecasting Using Machine Learning, 9 Months, 12/1/2026, 9/30/2027, 80000.00, Forecasting model developed using historical data and performance compared across test periods.

14. Cloud-Based Laboratory Equipment Monitoring Platform, 1 Year, 1/1/2025, 1/31/2026, 100000.00, Cloud dashboard developed with remote equipment status monitoring and notification capability.

15. Computer Vision Based Industrial Safety Compliance System, 18 Months, 2/1/2026, 8/31/2027, 65000.00, Inspection model trained and tested against sample safety scenarios with measurable detection results.

16. Intelligent Water Leakage Detection and Management, 2 Years, 3/1/2025, 3/31/2027, 55000.00, Leak detection prototype developed and tested using simulated pipeline leakage conditions.

17. Natural Language Processing Tool for Regional Language Learning, 6 Months, 4/1/2026, 10/31/2026, 72000.00, Language dataset prepared and a functional text-processing demonstration developed.

18. Smart Campus Navigation and Emergency Assistance System, 9 Months, 5/1/2025, 2/28/2026, 48000.00, Campus navigation prototype developed with mapped locations and emergency guidance workflow.

19. Predictive Maintenance Model for Industrial Motors, 1 Year, 6/1/2026, 6/30/2027, 88000.00, Predictive maintenance model developed and evaluated using historical equipment operating data.

20. AI Based Identification of Plastic Waste in Water Bodies, 18 Months, 7/1/2025, 1/31/2027, 62000.00, Image-based waste identification model developed and tested using representative water samples.

21. Development of a Solar Powered Cold Storage Prototype, 2 Years, 8/1/2026, 8/31/2028, 92000.00, Small-scale solar cold-storage prototype developed with temperature monitoring and performance testing.

22. Secure IoT Gateway for Industrial Sensor Networks, 6 Months, 9/1/2025, 3/31/2026, 58000.00, Secure gateway prototype developed with encrypted sensor communication and controlled access testing.

23. Data Analytics Platform for Small-Scale Manufacturing, 9 Months, 10/1/2026, 7/31/2027, 76000.00, Data analytics dashboard developed with sample manufacturing data and actionable performance indicators.

24. Automated Soil Nutrient Analysis and Recommendation System, 1 Year, 11/1/2025, 11/30/2026, 68000.00, Soil analysis workflow developed with a recommendation model and preliminary sample validation.

25. Edge AI Based Surveillance for Restricted Facilities, 18 Months, 12/1/2026, 6/30/2028, 105000.00, Edge AI prototype developed and evaluated for local processing of restricted-area monitoring events.

26. Low-Cost Flood Level Monitoring and Community Alert System, 2 Years, 1/1/2025, 1/31/2027, 47000.00, Flood-level sensing prototype developed with threshold-based alerts and simulated community notifications.

27. Digital Platform for Rural Entrepreneurship Support, 6 Months, 2/1/2026, 8/31/2026, 73000.00, Digital platform prototype developed with user workflows for registration, product discovery, and support.

28. Machine Learning Based Forecasting of Renewable Power Generation, 9 Months, 3/1/2025, 12/31/2025, 82000.00, Forecasting model developed and evaluated using historical renewable-generation datasets.

29. Smart Classroom Occupancy and Energy Optimization, 1 Year, 4/1/2026, 4/30/2027, 56000.00, Occupancy sensing prototype developed with automated energy-saving rules and controlled testing.

30. Development of a Portable Biomedical Signal Analysis Device, 18 Months, 5/1/2025, 11/30/2026, 98000.00, Portable signal-analysis prototype assembled and validated using representative biomedical signal samples.

31. AI Assisted Quality Inspection for Small Manufacturing Units, 2 Years, 6/1/2026, 6/30/2028, 69000.00, Automated inspection prototype developed and evaluated on representative manufacturing quality images.

Conference Details

1. uu, tu, sd, cdgf, tdr, International, UGC Care, 23, 2026-11, d, 6, Pending, 134

Publications

1. "hgdjffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff", jfh, hjh, N/A, pp. d, Oct 2026. DOI: h